Backend Assignment - JSON group by and Sort By Operators

Problem Statement

You are required to develop a Spring Boot application to handle JSON dataset records. The application should expose two APIs:

1. **Insert Record API**: To insert a JSON record into a specific dataset and persist it in the database.
2. **Query API**: To query a specific dataset and perform group-by and sort-by operations on the JSON records.

The application must:

- Use a relational database to store JSON dataset records.
- Provide clear request and response structures.
- Support dynamic group-by and sort-by operations based on the provided parameters.

Deliverables:

- A fully functional Spring Boot application with the above APIs.
- Source code hosted on a public Git repository.
- Documentation for the API endpoints and how to run the application.

Expected API Examples

Insert Record API
Request URL:
bash
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```
POST /api/dataset/{datasetName}/record
```

Request Body:
json
Copy code

```
{
    "id": 1,
    "name": "John Doe",
    "age": 30,
    "department": "Engineering"
}
```

Response:
json

Copy code

```
{
    "message": "Record added successfully",
    "dataset": "employee_dataset",
    "recordId": 1
}
```

1.

Query API with Group-By
Request URL:
bash
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```
GET /api/dataset/{datasetName}/query
```

Query Parameters:
makefile
Copy code

```
groupBy=department
```

Response:
json
Copy code

```
{
    "groupedRecords": {
        "Engineering": [
            { "id": 1, "name": "John Doe", "age": 30, "department":
"Engineering" },
            { "id": 2, "name": "Jane Smith", "age": 25,
"department": "Engineering" }
        ],
        "Marketing": [
            { "id": 3, "name": "Alice Brown", "age": 28,
"department": "Marketing" }
        ]
    }
}
```

2.

Query API with Sort-By
Request URL:
bash
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```
GET /api/dataset/{datasetName}/query
```

Query Parameters:
sql

Copy code

```
sortBy=age&order=asc
```

Response:

json

Copy code

```
{
    "sortedRecords": [
        { "id": 2, "name": "Jane Smith", "age": 25, "department":
"Engineering" },
        { "id": 3, "name": "Alice Brown", "age": 28, "department":
"Marketing" },
        { "id": 1, "name": "John Doe", "age": 30, "department":
"Engineering" }
    ]
}
```

Few things to take care of while submission of code:

- Assignment code submitted to be in Java
- The evaluation of the assignment will be based on code quality
- REST Compliant APIs, (No UI)
- Testable by Postman, (No UI)
- Will prefer classes properly refactored and following design patterns,
- Robustness in error handling
- Clarity of API documentation and how to run the project.
- Will Prefer TDD, JUnit